

03-09-2023

CHEMISTRY

QNO.

SECTION A

1.

4) Argon

2.

2) Calcium

3.

Molecular mass of Carbon dioxide = $12 + 32$

$$= 44 \text{ amu}$$

$$\text{Molecular mass} = 2 \times \text{VD}$$

$$44 \text{ amu} = 2 \times \text{VD}$$

$$\text{VD} = \frac{44 \text{ amu}}{2}$$

$$\text{VD} = 22$$

4) 22

4.

2) 17

5.

1g of H = 16g of O

2g of H = 32g of O (x 2 both sides)

the ratio $\Rightarrow \text{H}:\text{O} = 1:1$

3) 1:1

6.

4) Mercury

7.

2) Lead

8.

1) 1

9. 1) Lithium

10. 4) Ammonium chloride

11. 4) Copper chloride

13. 12, 3) An alkali

12.

12. 2) Consist of molecules

14. 1) direct combination

15. nitric acid and hydrochloric acid



16. i) a) The most reactive metal;
Y - Potassium

b) The most reactive non-metal
L - Fluorine

c) Family of L, O, R, T is Halogens

d) group 15 - Nitrogen
group 13 - Boron

e) Atomic radius decreases from left to right in periods
group 14 - C & group 16 - O So, O is smaller in radius

ii) a) reducing agent

b) bad

c) acidic

d) lead chloride

e) Chlorine : Cl

ii)

When gases react, they do so in volumes which bear a simple whole number ratio to one another and to the volume of the products, if gaseous, provided the volumes are measured under the same conditions of temperature and pressure. - Gay-Lussac's law

i) Molecular weight of $\text{SO}_2 = 32 + 2 \times 16$
 $= 64 \text{ amu}$

$$\text{number of moles} = \frac{\text{weight}}{\text{Molecular weight}}$$

$$= \frac{320}{64}$$

$$= 5 \text{ moles}$$

$$1 \text{ mole} = 22.4 \text{ litres (at STP)}$$

$$5 \text{ moles} = 22.4 \times 5$$

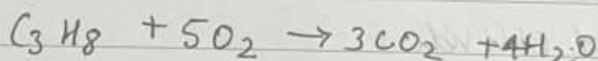
$$= 112 \text{ litres}$$

$\therefore$ Volume of SO_2 is 112 litres.

iii) $\text{Molecular weight} = 3 \times 12 + 8 = 36 + 8 = 44$

$$\text{Number of moles} = \frac{\text{weight}}{\text{Molecular weight}}$$

$$= \frac{8.8}{44} = 0.2 \text{ mole}$$



0.2 mole propane requires $\text{O}_2 =$

$$5 \times 22.4 (\text{litre}) \times 0.2$$

$$= 22.4 \times 1.0$$

$$= 22.4 \text{ litres}$$

iv. i) $\text{Number of gram atoms} = \frac{\text{weight}}{\text{Molecular weight}}$

$$= \frac{4.6}{23} = \frac{145}{230}$$

$$= 0.2 \text{ gram atoms}$$

ii) Molecular weight of $H_2O = 2 + 16 = 18 \text{ mol}$
 Molecular weight of $CuSO_4 \cdot 5H_2O = 64 + 32 + 64 + 90$
 $= 250 \text{ g mol}$
 $\% \text{ of } CuSO_4 \cdot 5H_2O = \frac{1}{5} \times 18 \times \frac{100}{250}$

$$= 36\%$$

iii) Empirical formula = Vapour density

$$\text{Molecular weight} = 2 \times \text{Vapour density}$$

$$\text{Molecular formula} = 2 \times XY_2$$

$$= X_2Y_4 //$$

vi. i) A solid non-metal belonging to the third period

— 15; P

ii) A metal of valency 1

— 19; K

iii) A gaseous element with valency 2

— 8; O

iv) An element belonging to group 2

— 4; Be

v) A inert gas

2; He

Section - B

1. i) Empirical formula = C_3H_4N

$$\text{Molecular weight} = 108$$

$$\text{Molecular weight of } C_3H_4N = 12 \times 3 + 4 \times 1 + 1 \times 14$$

$$= 36 + 4 + 14$$

$$= 40 + 14 = 54 \text{ amu}$$

$$\text{number of moles} = \frac{\text{weight}}{\text{molecular weight}}$$

$$= \frac{108}{54}$$

$$= 2 \text{ mole}$$

- ii) a. 1. Hydrochloric acid if added the pH will go down & it shows acidic, below 7 pH.
 2. A solution of base if added the pH will increase or will be the same but still shows pH above 7 i.e. alkali.

b. It belongs to group 15 or group VA.

iii) a) no. of moles of $\text{Ca}(\text{NO}_3)_2 = \frac{168}{84} = 2$

iv) a) no. of moles of $2\text{Ca}(\text{NO}_3)_2 = \frac{\text{weight}}{\text{molecular weight}}$

$$\text{Molecular weight} = 2[40 + 2(14 + 3 \times 16)]$$

$$= 2[40 + 2(14 + 48)]$$

$$= 2[40 + 2(62)]$$

$$= 2[40 + 124] = 2 \times 164 = 328$$

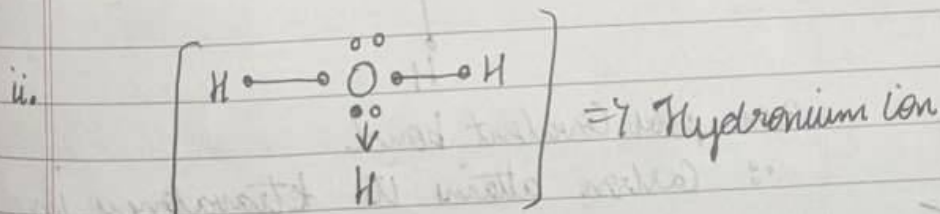
$$\text{No of moles} = \frac{8.2}{204}$$

$$\begin{aligned}\text{Molecular weight} &= 2 [40 + 2 (14 + 48)] \\ &= 2 [40 + 2 (62)] \\ &= 2 [40 + 124] \\ &= 2 (164)\end{aligned}$$

pH.

iv) is. When zinc and aluminium is heated with concentrated sodium hydroxide they liberate hydrogen with a pop sound and forming Sodium zincate and Sodium metakuminato.

b. b. When zinc reacts with ammonium hydroxide reaction forms chalky white precipitate and Calcium nitrate reacted with NH_4OH shows that Calcium nitrate is soluble in it.



2. i) 1. True, as noble gases will be stable so that it follows this stability

2. True; So α 's valency is 2 as $\text{P}_{\text{group}} 2$ electrons. It is good conductor of electricity as the forces are weaker in a molten state. Hence, low boiling point.

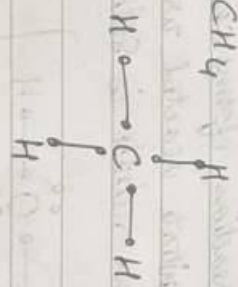
ii) mass of carbon = 12 gm of carbon atom

iii) a) its more than 1

b) its less than 1

c) X is to be placed to the left in the periodic table.

iv) 1. Methane CH_4



non-polar covalent bond

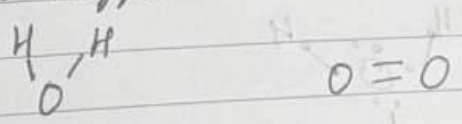
Carbon attains its tetravalency in methane.

ii. a)

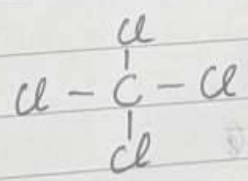
b) That these elements have 2 electrons left in their valency shell.

3. if a

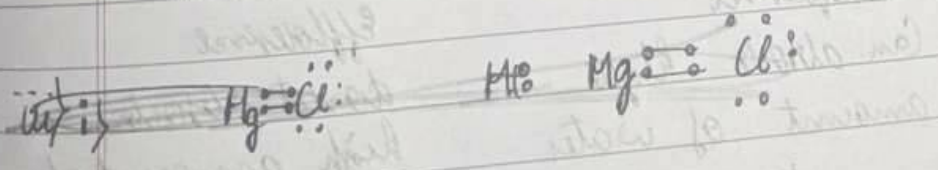
if at As we see that there are 2 single covalent bond between H & O. So, when O is Hydrogen needs only one electron to be stable but Oxygen needs 2 electron

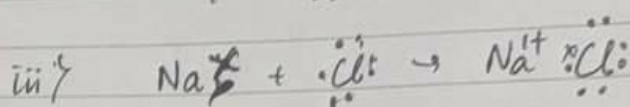
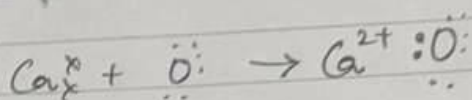
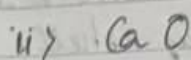
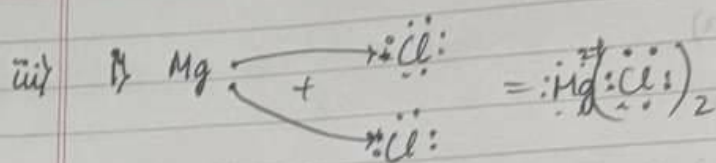


b) Carbon to attain its tetravalency



Chlorine needs only one e^- to be stable.



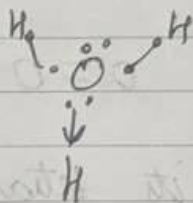


iv)

Conditions

• One lone pair of electrons must be present

• lone pair of electrons completely shared by donor



6. i) a. Copper sulphate

b. potassium nitrate

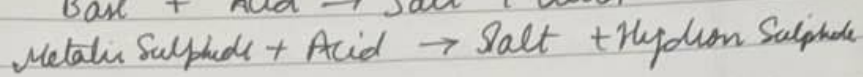
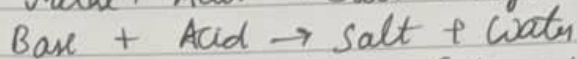
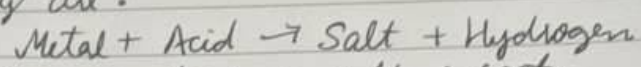
ii. Deliquescent

Can absorb high amount of water vapour

Efflorescent

do not absorb high amount of water vapour

iii) They are:



iv) a) H_3PO_4

- acidity

b) $\text{Ca}(\text{OH})_2$

- Basicity

c) NaOH

- Basicity

d) CH_3COOH

- Acidity

e) H_2SO_3

- Acidity

f) H_3PO_3

- Basicity.